

5.7.21. Suppose r is a fixed constant and $a_0, a_1, a_2, \dots$ is a sequence that satisfies the recurrence relation $a_k = ra_{k-1}$, for each integer $k \geq 1$ and $a_0 = a$. Use mathematical induction to prove that $a_n = ar^n$, for every integer $n \geq 0$.

Proof:

Suppose r is a fixed constant and $a_0, a_1, a_2, \dots$ is a sequence that satisfies the recurrence relation $a_k = ra_{k-1}$, for each integer $k \geq 1$ and $a_0 = a$.

We must show using induction that $a_n = r^n a = ar^n$, for every integer $n \geq 0$.

$$a_0 = a$$

$$a_1 = ra_0 = ra$$

$$a_2 = ra_1 = r(ra) = r^2a$$

$$a_3 = ra_2 = r(r^2a) = r^3a$$

$$a_4 = ra_3 = r(r^3a) = r^4a$$

Guess: $a_n = r^n a = ar^n$ for any arbitrary integer $n \geq 0$.

$$P(n) : a_n = r^n a = ar^n$$

Basic step: $P(0)$

$$P(0) : a_0 = r^0 a = ar^0 = a(1) = a$$

Therefore, $P(0)$ is true.

Inductive step: $P(k+1)$

Let $k \in \mathbb{Z}$, $k \geq 0$

Suppose $P(k)$ is true. Then:

$$P(k) : a_k = r^k a = ar^k$$

We must show that $P(k+1)$ is true:

$$P(k+1) : a_{k+1} = ar^{k+1}$$

From the recurrence relation:

$$a_{k+1} = ra_{(k+1)-1} = ra_k$$

By the inductive hypothesis:

$$a_{k+1} = r(ar^k) = ar^{k+1}$$

Therefore, $P(k+1)$ is true, which is what was to be shown.

Solution: Since $P(0)$ is true, and $P(k) \rightarrow P(k+1)$ is true, by proof by induction, $a_n = ar^n$, for every integer $n \geq 0$.